

Practice — Why Earth Is Unique: Its Position

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ**[1]**

The main reason Earth can support life is its —

- (a) blue colour (b) right distance from the Sun (c) number of moons (d) fast rotation

Q2 · MCQ**[1]**

The range of distances where liquid water can exist is the —

- (a) ozone layer (b) habitable zone (c) atmosphere (d) magnetic field

Q3 · MCQ**[1]**

The hottest planet in the solar system is —

- (a) Mercury (b) Venus (c) Mars (d) Jupiter

Q4 · MCQ**[1]**

Earth is called the Blue Planet because —

- (a) the sky is blue (b) about 70% of its surface is water (c) it is cold (d) of its atmosphere only

Q5 · ASSERTION-REASON**[1]**

A: Venus is hotter than Mercury.

R: Venus's thick carbon dioxide atmosphere traps heat.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

Gases like CO₂ trapping the Sun's heat is called the _____ effect.

Q7 · FILL IN THE BLANK

[1]

For life as we know it, _____ water is essential.

Short answer (2 marks each)

Q8

[2]

Why is Earth's distance from the Sun ideal for life?

Q9

[2]

What is the habitable zone?

Q10

[2]

What is the greenhouse effect and why does it matter for Earth?

Long / application (3 marks each)

Q11

[3]

Explain why Earth is habitable but Venus is not, comparing their temperatures and atmospheres.

Q12

[3]

A spaceship carries soil and water to a planet far beyond the habitable zone. Would plants grow there? Explain.

[Check yourself](#) → Answer key